

Golden Men's Fashion

An E-Commerce Web Application for Premium Menswear Retail

Project Report — EWA408510: E-Commerce and Web Application

1. Introduction

Golden Men's Fashion is an online storefront built for a fictional Rwandan menswear retailer that wants to move its catalogue and sales process onto the web. The platform lets visitors browse a curated collection of suits and accessories, view product details, and reach the business through a contact form, while an authenticated dashboard gives the store administrator a quick view of inventory, users, and orders. The system was designed, built, and deployed end to end as the final project for the EWA408510 E-Commerce and Web Application course.

2. Problem Statement

Many small and medium menswear retailers in Rwanda still rely entirely on physical stores and informal social media posts to reach customers. This limits their visibility, makes it difficult for customers to browse a full catalogue before visiting, and leaves the business with no structured way to record products, customers, or orders. Golden Men's Fashion addresses this gap by providing a dedicated web platform where the catalogue, customer interactions, and order data are organized, searchable, and accessible from any device.

3. Objectives

- Provide a responsive, professional storefront where customers can browse and search products by category.
- Allow customers to view detailed product information, including price and category, before purchasing.
- Give the business owner a secure dashboard to monitor products, users, and orders in one place.
- Store product, user, and order data reliably in a relational database.
- Demonstrate a complete, modern software delivery workflow: version control, containerization, and automated deployment.

4. System Features

4.1 Public-Facing Site

- Home page introducing the brand with a clear call to action to shop the collection.
- About page describing the company's story, mission, vision, and key performance figures.
- Shop page listing all products with a search bar and category labels.
- Contact page with a message form and the business's location, phone, and hours.

4.2 Admin Dashboard

- Login-protected dashboard summarizing total users, total products, total orders, and revenue.
- Quick actions for adding products, viewing orders, and managing registered users.

5. Technologies Used

Layer	Technology
Frontend	React (deployed on Netlify)
Backend / API	Python with Flask
Database	MySQL
Hosting (Backend & Database)	Railway
Hosting (Frontend)	Netlify
Version Control	Git and GitHub

6. System Architecture

The application follows a three-tier architecture. The React frontend, hosted on Netlify, handles all presentation and user interaction and communicates with the backend over HTTPS using REST API calls. The Flask backend, hosted on Railway, exposes the API endpoints, applies business logic, and manages authentication for the dashboard. The MySQL database, also hosted on Railway, stores all persistent data covering products, users, and orders. Separating these three layers keeps the frontend independently deployable, allows the backend to scale or be modified without affecting the user interface, and keeps the data layer centralized and secure.

- Client Layer: React single-page application (Netlify).
- Application Layer: Flask REST API handling requests, validation, and authentication (Railway).
- Data Layer: MySQL relational database storing products, users, and orders (Railway).

7. Screenshots

The following screenshots show the core pages of the deployed application, captured from an authenticated session.



Figure 1: Home page

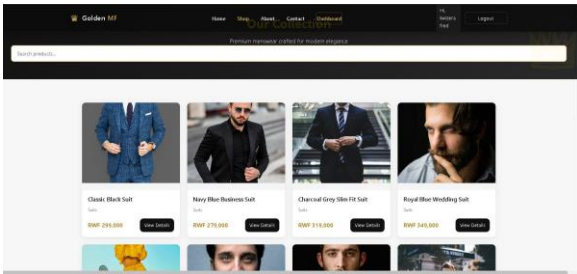


Figure 2: Shop / product listing

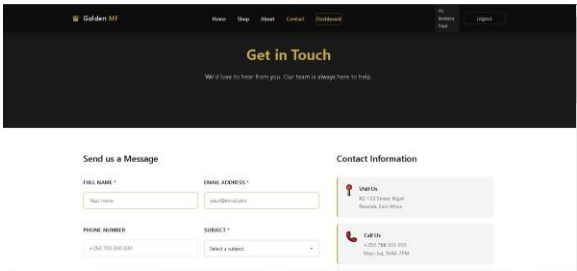
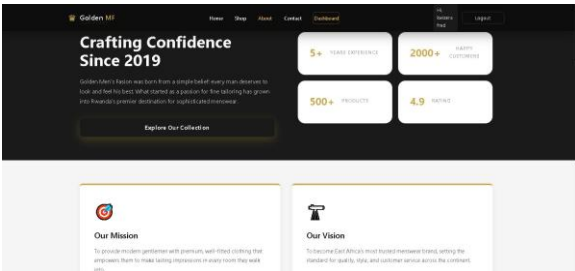


Figure 3: About page

Figure 4: Contact page

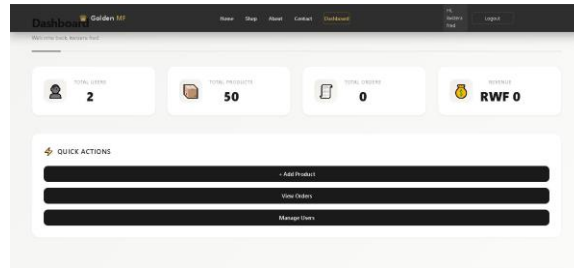


Figure 5: Admin dashboard

8. GitHub Repository and Deployment Links

GitHub Repository: <https://github.com/KWIZERA-FRED/Golden-Men-s-Fashion.git>

Live Deployment URL: <https://goldenmensfashion.netlify.app/>

9. CI/CD Description

The frontend is connected directly to Netlify's continuous deployment pipeline: every commit pushed to the main branch of the GitHub repository triggers an automatic build of the React application and a redeploy of the live site, so the production URL always reflects the latest committed code. The backend and database are hosted on Railway, which similarly redeploys the Flask service whenever changes are pushed to the connected repository branch, rebuilding the application container and restarting the service with the updated code. Together, these two managed pipelines give the project an automatic build-and-deploy flow on every push without requiring a separately written workflow file.

10. Challenges Encountered

- Coordinating environment variables and connection strings between the Netlify frontend and the Railway-hosted Flask API and MySQL database.
- Handling cross-origin requests (CORS) between the Netlify domain and the Railway API domain.
- Structuring the MySQL schema to cleanly support products, users, and orders while keeping queries simple for the Flask API.
- Keeping the dashboard's statistics (users, products, orders, revenue) in sync with live database values.

11. Future Work

- Integrate a mobile money or card payment gateway to support real online checkout.
- Add a customer-facing order history and tracking page.
- Introduce product reviews and ratings to build customer trust.
- Add an analytics dashboard with sales trends over time.

12. Conclusion

Golden Men's Fashion demonstrates a complete e-commerce workflow, from a responsive React storefront through a Flask API to a MySQL database, deployed across Netlify and Railway with automatic redeployment on every push. The project satisfies the core requirements of the EWA408510 final examination, including a

professional user interface, product browsing, an administrative dashboard, database integration, GitHub version control, and live deployment, while leaving a clear path for future enhancements such as payments and analytics.